

ClearPath Child Review (CCR)

Developer Implementation Specification

Version: 1.0 | Project: LMS Integration

1. System Architecture Overview

The CCR is a multi-respondent 360° assessment tool. The system must collect data from four primary roles and aggregate them into a single **Student Growth Index (SGI)**.

- **Primary Key:** `student_id`
- **Respondent Prefixes:** `S_` (Student), `F_` (Father), `M_` (Mother), `T_` (Teacher).
- **Output:** Automated PDF Report with Radar Chart and SGI Scores.

2. Database Schema (Suggested)

```
TABLE ccr_responses (  
    response_id INT PRIMARY KEY AUTO_INCREMENT,  
    student_id INT NOT NULL, -- Links to LMS student record  
    academic_session_id INT,  
    term INT,  
    respondent_type ENUM('student', 'father', 'mother',  
    'teacher'),  
    data JSON, -- Stores question_id: value pairs  
    comments TEXT, -- For the "One thing the school should  
    know" question  
    is_submitted BOOLEAN DEFAULT FALSE,  
    created_at TIMESTAMP DEFAULT CURRENT_VALUE
```

);

3. Weighting & Scoring Logic

The SGI is a weighted average across 10 Domains. The "Domain Authority" weights are as follows:

Domain	Student (S_)	Father (F_)	Mother (M_)	Teacher (T_)
Identity	60%	10%	10%	20%
Academic	20%	15%	15%	50%
Character	30%	17.5%	17.5%	35%
Emotional	70%	10%	10%	10%
Social	20%	10%	10%	60%

SGI Formula:

$$\text{SGI score} = (\text{Foundation} * 0.35) + (\text{Performance} * 0.35) + (\text{Environment} * 0.15) + (\text{Aspiration} * 0.15)$$

4. Reliability & Red Flags

Conflict Detection: If $\text{ABS}(\text{S_Score} - \text{T_Score}) > 40\%$, flag as "Observation Gap" in the PDF.

Red Flag Triggers:

- IF $\text{S_Happiness} < 2$ AND $\text{S_Loneliness} > 4$ -> Trigger "Immediate Counselor Alert".
 - IF $\text{S_Safety} < 2$ -> Trigger "Safety Safeguarding Alert".
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5. Questionnaire Structure (Sample JSON)

```
{
  "student_id": "123456",
  "responses": {
    "S_Q1": 5, // Strongly Agree
    "S_Q2": 4,
    "F_Q1": 3, // Father's observation
    "M_Q1": 4, // Mother's observation
    "T_Q1": 5 // Teacher's observation
  },
  "innovation_question": "He has been very anxious about the
upcoming divorce."
}
```

6. PDF Report Visualization Requirements

- **Cover Page:** Student Name, `student_id`, and **Student Photo** (fetched from LMS path).
- **Radar Chart:** 10 axes (one for each domain). Overlay Student vs. Aggregate Adult scores.
- **Prescription Box:** Logic-driven text based on low scores.
Example: If Social < 50, output "Encourage participation in team-based extracurriculars."
- **The "Human Insight" Box:** Display the answer to the final innovation question prominently on page 2.